

23CSE202 DATABASE MANAGEMENT SYSTEMS

S3 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

PROJECT ABSTRACT

Group No : C8

Team Members :

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Project Title: Online Movie Ticket Booking System

Abstract:

Real World Problem

We've all experienced the frustration of trying to book a movie ticket only for the app to freeze, or worse, finding out a seat was double-booked. While buying a ticket online seems simple to a user, managing the data behind the scenes during a blockbuster release requires a highly efficient database.

Objectives

The core goal is to design a clean, normalized relational database that seamlessly links theatres, screens, movies, showtimes, seats, and customers. The system focuses on real-time seat tracking to eliminate double-bookings using concurrency control, handling secure transactions and cancellations, and providing admins with clear insights into revenue and audience trends.

Proposed database system

Model: Relational model

Relations:

- Theatre: Stores theatre details like location, name, contact information and number of screens. Parent to all screens that belong to the theatre.
- Movie: Stores movie details like title, genre, language, duration, image url, and release date. It connects to Showtime.
- Screen: Stores each screen details like seating capacity, screen number and theatre. It is the child of theatre.
- Seat: Lists seats related to a screen with type and availability.
- Customer: Stores customer login details and connects to individual customer booking history.

- **Showtime:** Links the movies and the screen, used to specify date, time and price tier.
- **Price:** Stores price tier and price of each seat according to day, time and seat type. Related to seat and Showtime.
- **Booking:** Records which customer reserved which seat for which show along with booking status, and time. Relates Customer, Seat, and Showtime.
- **Cancellation:** Stores cancelled bookings, refund status, and reason.
- **Payment:** Records payments and transaction details with details like amount, method, status.

Intended Users

The system is designed for three distinct groups:

1. **Customers** (who browse, choose seats, and book tickets)
2. **Theatre Administrators** (who manage show schedules, screen layouts, and pricing).
3. **System Administrators** (who oversee the entire platform, manage accounts, and pull analytical reports).

Major Functionalities

The system will feature secure user registration and login, along with a searchable movie catalogue filtered by genre, date, or theatre. Customers can view an interactive seat layout, select their seats, and generate a basic digital ticket booking. On the administrative side, the database will support tools for managers to easily add or update showtimes, adjust ticket pricing based on seat types, process cancellation, and view simple summaries of total tickets sold and total revenue per movie.